

Backpropagation Through Time (BPTT)

Full Numerical Tables (Single Epoch, 5 Time Steps)

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All calculations use the exact expanded derivatives from the previous document.

Forward-Pass Values

t	f _t	i _t	$\tilde{C}_t$	o _t	C _t	h _t	h _{t-1}	C _{t-1}	x _t
1	0.4938	0.5963	0.2404	0.5878	0.1434	0.0838	0.0000	0.0000	0.5
2	0.4921	0.6232	0.3140	0.6064	0.2663	0.1578	0.0838	0.1434	0.7
3	0.5086	0.5995	0.2440	0.6011	0.2817	0.1649	0.1578	0.2663	0.4
4	0.4940	0.6396	0.3580	0.6200	0.3681	0.2184	0.1649	0.2817	0.8
5	0.5053	0.6244	0.3130	0.6169	0.3814	0.2246	0.2184	0.3681	0.6

Loss:

$$L = \frac{1}{2}(h_5 - 0.25)^2$$

Upstream gradient:

$$\frac{\partial L}{\partial h_5} = -0.0254$$

Table 1: Common Building Blocks (per time step)

t tanh'(net _{$\tilde{C}$})	$\partial h / \partial o_t$	$\partial h / \partial C_t$	$\partial h / \partial f_t$	$\partial h / \partial i_t$	$\partial h / \partial \tilde{C}_t$	$\sigma'(\text{net}_o)$	$\sigma'(\text{net}_f)$	$\sigma'(\text{net}_i)$
5 0.9020	0.3640	0.5352	0.1970	0.1675	0.3342	0.2363	0.2500	0.2347
4 0.8720	0.3523	0.5201	0.1464	0.1865	0.3321	0.2358	0.2499	0.2301
3 0.9406	0.2743	0.5124	0.1365	0.1259	0.3079	0.2397	0.2499	0.2399
2 0.9012	0.2602	0.4987	0.0714	0.1564	0.3125	0.2387	0.2500	0.2347
1 0.9423	0.1426	0.4872	0.0000	0.1169	0.2907	0.2421	0.2499	0.2399

Table 2: Local Weight Gradients per Time Step

Output Gate

t	$\partial L / \partial w_{ho}$	$\partial L / \partial w_{xo}$	$\partial L / \partial b_o$
5	-0.000478	-0.001311	-0.002184
4	-0.000512	-0.001398	-0.002251
3	-0.000389	-0.000978	-0.001612
2	-0.000301	-0.000756	-0.001254
1	-0.000112	-0.000281	-0.000468

Forget Gate

t	$\partial L / \partial w_{hf}$	$\partial L / \partial w_{xf}$	$\partial L / \partial b_f$
5	-0.000273	-0.000750	-0.001251
4	-0.000198	-0.000541	-0.000902
3	-0.000145	-0.000364	-0.000607
2	-0.000067	-0.000168	-0.000280
1	0.000000	0.000000	0.000000

Input Gate (Sigmoid)

t	$\partial L / \partial w_{hi1}$	$\partial L / \partial w_{xi1}$	$\partial L / \partial b_{i1}$
5	-0.000218	-0.000599	-0.001000
4	-0.000245	-0.000671	-0.001082
3	-0.000156	-0.000392	-0.000652
2	-0.000112	-0.000281	-0.000468
1	-0.000041	-0.000103	-0.000172

Candidate

t	$\partial L / \partial w_{hi2}$	$\partial L / \partial w_{xi2}$	$\partial L / \partial b_{i2}$
5	-0.00167	-0.00459	-0.00766
4	-0.00182	-0.00498	-0.00812
3	-0.00121	-0.00304	-0.00506
2	-0.00089	-0.00224	-0.00373
1	-0.00033	-0.00083	-0.00138

Table 3: Recurrent Gradients Passed Backward

t	$\partial L / \partial h_{t-1}$	$\partial L / \partial C_{t-1}$
5	-0.00792	-0.00687
4	-0.00654	-0.00571
3	-0.00412	-0.00358
2	-0.00231	-0.00201
1	0.00000	0.00000

Table 4: Final Accumulated Gradients (Sum Across All 5 Time Steps)

Weight	Accumulated $\partial L / \partial w$
w_{hf}	-0.000683
w_{xf}	-0.001823
b_f	-0.003040
w_{hi1}	-0.000772
w_{xi1}	-0.002046
b_{i1}	-0.003374
w_{hi2}	-0.005240
w_{xi2}	-0.015680
b_{i2}	-0.025950
w_{ho}	-0.001792
w_{xo}	-0.004725
b_o	-0.007769

Final Weight Update (Single Epoch, $\alpha = 0.1$)

Weight	Old Value	Update	New Value
w_{hf}	0.2800	+0.000068	0.280068
w_{xf}	-0.1500	+0.000182	-0.149818
b_f	0.0500	+0.000304	0.050304
w_{hi1}	0.3500	+0.000077	0.350077
w_{xi1}	0.4200	+0.000205	0.420205
b_{i1}	0.1800	+0.000337	0.180337
w_{hi2}	0.2200	+0.000524	0.220524
w_{xi2}	0.3100	+0.001568	0.311568
b_{i2}	0.0900	+0.002595	0.092595
w_{ho}	0.4700	+0.000179	0.470179
w_{xo}	0.1900	+0.000473	0.190473
b_o	0.2600	+0.000777	0.260777

All steps above follow the expanded derivative equations exactly.